Chapter 18, Problem 22P

Problem

Prove that if $F(\omega)$ is the Fourier transform of $f(t)$, $\mathcal{F}[f(t)\sin\omega_0 t] = \frac{j}{2}[F(\omega+\omega_0) - F(\omega-\omega_0)]$

Step-by-step solution

Step 1 of 1

we have

$$F(\omega) = F\left[f(t)\right]$$

$$\Rightarrow F\left[f(t)\sin\omega_o t\right]$$

$$= F\left[f(t)\left\{\frac{e^{j\omega_o t}}{2s}\right\}\right]$$

$$= \frac{1}{2j}F\left[f(t)e^{j\omega_o t} - f(t)e^{-j\omega_o t}\right]$$

$$= \frac{1}{2j}\left[F\left[f(t)e^{j\omega_o t}\right] - F\left[f(t)e^{-j\omega_o t}\right]\right]$$

$$= \frac{1}{2j}\left[F(\omega-\omega_o) - F(\omega+\omega_o)\right] \qquad \left[j^2 = -1\right]$$

$$= \frac{j}{2}\left[F(\omega+\omega_o) - F(\omega-\omega_o)\right]$$